Beyond the VAE Bottleneck: One-Step Pixel Diffusion from Latent Teachers

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Abstract

One-step text-to-image diffusion reduces generation to a single denoising pass, but most existing models operate in latent space and therefore require VAE decoding. In this regime, the decoder becomes a non-negligible inference bottleneck, often comparable to the cost of the denoising network itself. Pixel diffusion removes this bottleneck and offers a simpler generation pipeline, yet *high-quality one-step pixel generation* remains underexplored due to the lack of strong pixel-space teachers. In this work, we study **heterogeneous score distillation**: distilling a one-step pixel student from a latent diffusion teacher. A naive solution projects the student's pixel outputs into the teacher's latent space and applies standard distribution matching distillation. We demonstrate that this design is limited by three factors: VAE-induced loss of fine-grained details, insufficient semantic supervision from existing VFM-based adversarial objectives, and spectral mismatch that disrupts the coarse-to-fine learning dynamics. To address these issues, we introduce a heterogeneous distillation framework with four key components: (1) a **student-based pixel adversarial objective** for VAE-free detail supervision while reducing training latency, (2) a **decoupled VFM adversarial objective** that separates semantic alignment from structural realism, (3) **curriculum-driven, spectrum-aware objectives** in both pixel and score spaces that encourage coarse-to-fine training, and (4) a **preference-calibrated model merging strategy** that combines stronger human-preference alignment with gains in compositional and fine-grained grounding. Extensive experiments show that our one-step pixel diffusion achieves a HPSv2 score of **34.32**, surpassing strong baselines on key preference metrics while remaining competitive across compositional and grounding metrics, while delivering at least $\mathbf{1.5}\times$ faster inference than other one-step latent diffusion.

1 Introduction

Text-to-image diffusion models [16, 9, 2] have achieved remarkable progress in visual fidelity, prompt alignment, and compositional generation, largely driven by scalable latent diffusion models (LDMs). However, standard sampling still requires many denoising steps, making high-quality generation slow and computationally expensive. Diffusion distillation addresses this by compressing multi-step teachers into one- or few-step generators. Existing approaches include consistency-based [34, 33, 7, 18, 11, 10, 22], adversarial [30, 29], progressive [28], score-based [46, 23, 8, 43, 42, 4, 19], and hybrid methods [6]. Among them, score-based methods remain dominant, consistently achieving strong performance across architectures [27, 25, 9, 16].

Despite this progress, most one-step text-to-image generators operate in latent space and require a variational autoencoder (VAE) decoder to reconstruct RGB images. This design is well suited to multi-step sampling, where tens of denoising evaluations dominate the total inference cost. However, in the one-step setting, the balance changes: after the denoising cost has been largely removed, VAE

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decoding. Moreover, VAE reconstruction is lossy and can attenuate fine-grained textures, making the final image depend not only on the denoising model but also on the VAE decoder.

Pixel diffusion addresses this remaining inefficiency by generating directly in RGB space, removing the VAE decoder from inference. Although early pixel-space models [17] lagged behind latent diffusion, recent models [21, 37] have narrowed the quality gap, making pixel diffusion increasingly attractive for one-step generation, where decoder removal directly improves throughput. However, one-step pixel diffusion remains underexplored because distillation is teacher-limited: the teacher defines the target distribution and largely sets the student's quality ceiling. Strong pixel-space teachers are thus essential, yet existing pixel diffusion models still lag behind modern LDMs in scale, data, and generative quality. Existing pixel diffusion models [21, 37] are around 1B parameters, whereas recent LDMs [25, 16, 9, 2] scale to 4–12B parameters and benefit from substantially larger training compute and data. Therefore, training a pixel teacher of comparable quality and size is prohibitively computationally expensive and time-consuming.

This raises a natural but largely unexplored question: *Can we bypass the need for an expensive pixel-space teacher by transferring the rich generative priors of a powerful LDM into a one-step pixel generator?* We study this by extending Distribution Matching Distillation to a cross-representation setting, which we call **heterogeneous score distillation**: latent-space teacher scores supervise pixel-space outputs from a one-step generator. A natural approach is to encode generated images with the teacher's VAE and apply standard latent-space distillation. However, this bridge has three limitations. First, the latent teacher is blind to fine-grained pixel details discarded by the VAE encoder, leaving these textures having no supervision. Second, image-only VFM [24, 15, 3] adversarial objectives used in prior distillation methods [4, 19] improve realism but provide little direct feedback on text alignment. Third, autoencoders can introduce spectral bias in pixel–latent mappings by amplifying high frequencies in latent components [31]. This disrupts coarse-to-fine learning and can cause the student to overfit premature high-frequency signals.

By explicitly identifying these limitations, we design an enhanced heterogeneous distillation pipeline that bridges the gap between latent-space priors and pixel-space synthesis. Specifically, we propose:

1. **Student-Based Pixel Adversarial Distillation:** We introduce a pixel-domain adversarial objective initialized from the lightweight student rather than the latent teacher. This reduces training latency while bypassing VAE's lossy compression, enabling sharper synthesis.
2. **Semantic VFM-based Adversarial Distillation**: To address the inadequate semantic guidance, we propose an adversarial loss that operates across various foundation encoders, combining their benefits for accurate semantic supervision and enhanced realism.
3. **Spectral Curriculum Objectives**: We identify a frequency mismatch between pixel and latent representations that can cause latent-space supervision to over-emphasize amplified high-frequency information. Hence, we introduce a spectral curriculum into two objectives: a scheduler-weighted spectrum-profile loss and a spectrum-weighted DMD objective. Together, they prioritize low-frequency structure in early training iterations, progressively introduce controlled high-frequency details, and improve training stability.
4. **Preference-Calibrated Model Merging**: To further improve alignment with human preferences, we merge the checkpoint trained with all aforementioned objectives with a preference-tuned checkpoint optimized using score distillation and a bounded preference objective. This combines the human-preference improvements of preference tuning with the fine-grained grounding capabilities of the full distillation model.

With these components, we effectively adapt the score distillation framework to pixel diffusion models. To the best of our knowledge, this is the first systematic study of distilling a high-quality one-step pixel diffusion model from latent teachers. Extensive experiments show that our model achieves a state-of-the-art HPSv2 score of **34.32**, surpassing existing multi-step pixel diffusion models as well as many latent diffusion baselines, both one-step and multi-step, while delivering at least $\mathbf{1.5}\times$ faster inference in the one-step regime.

2 Preliminaries

2.1 Text-to-Image Diffusion Models

Diffusion models [13, 32, 35] learn to reverse a noise-corruption process from clean data $x_0 \sim p_{\text{data}}$ to Gaussian noise, conditioned on text prompt y in text-to-image generation.

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Latent diffusion models. Latent diffusion models (LDMs) [27, 25, 9, 16, 2] encode an RGB image into a latent $z_0 = \mathcal{E}(x_0)$, perform diffusion in latent space, and decode with $\mathcal{D}$. The latent forward process is defined as:

$$q(z_t|z_0) = \mathcal{N}(\alpha_t z_0, \sigma_t^2 \mathbf{I}), \qquad z_t = \alpha_t z_0 + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}), \tag{1}$$

where $\alpha_t = \sqrt{\bar{\alpha}_t}$ and $\sigma_t = \sqrt{1 - \bar{\alpha}_t}$. A noise estimator ϵ_θ is trained with the standard objective: $\min_\theta \mathbb{E}\|\epsilon - \epsilon_\theta(z_t, t, y)\|_2^2$ (2). At inference, the clean latent is decoded as $x_0 = \mathcal{D}(z_0)$.

Flow matching. Recent large-scale models [9, 16, 2] often use flow matching, which defines a linear interpolation between data z_0 and noise ϵ:

$$z_t = (1-t)z_0 + t\epsilon, \quad v = \epsilon - z_0, \quad \hat{z}_0 = z_t - t\, v_\theta(z_t, t, y), \qquad t \in [0, 1]. \tag{3}$$

The velocity model is trained by minimizing the objective $\mathbb{E}\|v_\theta(z_t, t, y) - (\epsilon - z_0)\|_2^2$ (4).

2.2 Pixel Diffusion Models

Pixel diffusion models generate directly in RGB space without a VAE. Instead of diffusing latents, they apply the forward process directly to images, $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sigma_t \epsilon$, where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ and $\sigma_t = \sqrt{1 - \bar{\alpha}_t}$. The model predicts noise or velocity in pixel space, so inference directly outputs RGB images without VAE decoding, making this paradigm appealing for one-step generation.

2.3 Distribution Matching Distillation

Distribution Matching Distillation (DMD) [43, 42] distills a one-step generator by minimizing the reverse KL divergence between the student distribution p_{fake} and teacher distribution p_{real}. Given G_θ, input noise $\xi \sim \mathcal{N}(0, I)$, and generated sample $z_0 = G_\theta(\xi, y)$, we obtain z_t using the teacher forward process. The reverse-KL gradient is

$$\nabla_\theta \mathcal{L}_{\text{KL}} = -\mathbb{E}_{\xi, t, z_t}\left[\left(s_{\text{real}}(z_t, t, y) - s_{\text{fake}}(z_t, t, y) \right) \frac{dG_\theta(\xi, y)}{d\theta} \right], \tag{5}$$

where $s_{\text{real}} = \nabla_{z_t} \log p_{\text{real}}(z_t)$ and $s_{\text{fake}} = \nabla_{z_t} \log p_{\text{fake}}(z_t)$ are approximated by a pretrained teacher T_ϕ and a fake-score network T_ψ, respectively.

DMD2 [42] further adds a latent-space adversarial objective:

$$\mathcal{L}_{\text{GAN}}^{DMD2} = \mathbb{E}_{z_{\text{ref}} \sim p_{\text{real}}, t, y}\left[\log D(z_{\text{ref},t}, t, y)\right] + \mathbb{E}_{z \sim \mathcal{N}(0,I),\, z = G_\theta(z), t, y}\left[-\log D(z_t, t, y)\right], \tag{6}$$

where D is derived from T_ψ with additional trainable heads, and z_{ref} denotes real data represented in latent space. We use DMD and DMD2 interchangeably throughout the paper; both refer to DMD2.

3 Method: Enhanced Heterogeneous Score Distillation for Pixel Diffusion

We extend heterogeneous score distillation to better constrain RGB-space supervision when training a one-step pixel student from a latent teacher. Beyond the naive latent-to-pixel formulation in Sec. 3.1, our framework adds four components: a student-based pixel GAN for fine-grained details (Sec. 3.2), a decoupled VFM adversarial loss for semantic and structural guidance (Sec. 3.3), spectral curriculum objectives for coarse-to-fine learning (Sec. 3.4), and preference-calibrated model merging for improved human preference alignment (Sec. 3.6).

3.1 Problem Setup: Heterogeneous score distillation

We define **heterogeneous score distillation** as transferring latent-space teacher scores to a pixel-space student. Let T_ϕ be a pretrained multi-step latent teacher and G_θ a one-step pixel generator. Given prompt y and Gaussian input $\xi \sim \mathcal{N}(0, I)$ in RGB space, the student directly predicts

$$x_0 = G_\theta(\xi, y), \qquad \xi, x_0 \in \mathbb{R}^{3 \times H \times W}. \tag{7}$$

Since the teacher operates in latent space while the student outputs pixels, we bridge them with the frozen VAE encoder $\mathcal{E}$: $z_0 = \mathcal{E}(x_0) = \mathcal{E}(G_\theta(\xi, y))$. The encoded sample z_0 is then used in the DMD objectives in Eq. 5 and Eq. 6. This simple latent-to-pixel baseline exposes the bottlenecks of cross-representation distillation and motivates our proposed components.

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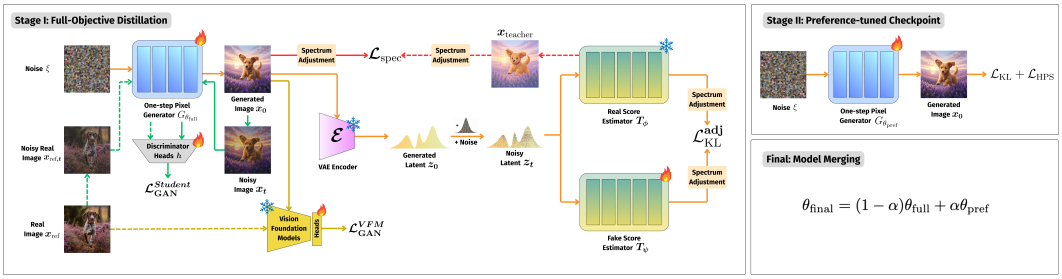

Figure 1: **Overview of our heterogeneous score distillation framework. Stage I** trains a one-step pixel generator $G_{\theta_{\text{full}}}$ that directly produces an RGB image x_0. The generated image is projected to the teacher latent space by the frozen VAE encoder $\mathcal{E}$ for latent score distillation, while pixel-space supervision is provided by a student-based adversarial loss $\mathcal{L}_{\text{GAN}}^{\text{Student}}$, a decoupled VFM adversarial loss $\mathcal{L}_{\text{GAN}}^{\text{VFM}}$, and spectral curriculum objectives ($\mathcal{L}_{\text{spec}}$ and $\mathcal{L}_{\text{KL}}^{\text{spec}}$). **Stage II** trains a preference-tuned checkpoint $G_{\theta_{\text{pref}}}$ with $\mathcal{L}_{\text{KL}} + \mathcal{L}_{\text{HPS}}$.The final model is obtained by linearly merging the full-objective and preference-tuned checkpoints. *(Zoom in for details.)*

3.2 Pixel-Space Adversarial Design

While the GAN objective in DMD2 (Eq. 6) significantly improves one-step generation, naively adopting it for our heterogeneous score distillation is suboptimal. Since the discriminator is based on the latent fake score estimator T_ψ, this approach introduces two critical bottlenecks:

1. **Information Loss via VAE Encoding.** Projecting pixels into the latents via $\mathcal{E}$ reintroduces a limitation of LDMs: the loss of fine-grained details [44]. Even strong VAEs [5, 16] attenuate sharp details during encoding, limiting the fidelity of generation. This lossy projection weakens supervision: since the teacher and discriminator only observe the VAE-encoded latent $\mathcal{E}(x) \in \mathcal{Z}$, pixel details discarded by $\mathcal{E}$ are absent from the teacher scores and discriminator feedbacks. As a result, the student receives no signal to recover these details.
2. **Training Latency.** In DMD2, the discriminator D is built by attaching a classification branch to the fake score estimator T_ψ and reusing its encoder features up to the bottleneck. While feasible for smaller models, this design becomes slow for modern billion-parameter diffusion backbones. Since T_ψ is already a trainable model comparable in size to the teacher, using it as the discriminator backbone requires extra forward and backward passes through a teacher-scale discriminator at each update, substantially increasing training latency.

Student-based Pixel Adversarial Objectives. To overcome these structural bottlenecks, we decouple the discriminator entirely from the latent teacher. Following [40], using an online network as the adversarial feature extractor complicates optimization, since the same network must balance score matching and GAN discrimination. Instead, we initialize D_{student} from a pretrained multi-step *pixel*-space diffusion model with lightweight trainable heads, similar to [29]. Because D_{student} operates directly in RGB space, it bypasses the VAE projection $\mathcal{E}$ and can penalize fine-grained details that are invisible in the latent space $\mathcal{Z}$. As shown in fig. 2a, this yields sharper textures, significantly less artifacts, and more detailed local structures.

As shown in fig. 2b, the total per-iteration latency drops from 4998.4ms to 2514.1ms. The slightly higher peak memory comes from full-resolution RGB activations, but this trade-off substantially reduces training time while providing direct pixel-space adversarial feedback. Formally, given a text prompt y, a generated image $x_0 = G_\theta(\xi, y)$, and a real image $x_{\text{ref}} \sim p_{\text{data}}$, we train the discriminator D_{student} with:

$$\mathcal{L}_{\text{GAN}}^{Student} = \mathbb{E}_{x_{\text{ref}} \sim p_{\text{data}},\, t,y}\left[f\left(D_{\text{student}}(x_{\text{ref},t}, t, y)\right)\right] + \mathbb{E}_{\xi \sim N(0,I), y, x_0 = G_\theta(\xi,y),\, t}\left[f\left(-D_{\text{student}}(x_t, t, y)\right)\right], \tag{8}$$

where $f(t) = -\log(1 + e^{-t})$ denotes the Softplus function, and $x_{\text{ref},t}$ and x_t are the noisy versions of the real image x_{ref} and generated image x_0 at timestep t, respectively.

3.3 Decoupled VFM Adversarial Supervision

Pixel-space generation enables direct supervision from Vision Foundation Models (VFMs), avoiding the need to decode latent samples. Existing VFM-based adversarial objectives [4, 19] typically rely

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(a) Qualitative comparison

(b) Training efficiency at 512×512

Method	Total ↓ ms/iter	Fwd. ↓ ms	Bwd. ↓ ms	Alloc. ↓ GB	Reserv. ↓ GB
KL + Teacher GAN	4998.4	2261.4	2728.5	**60.0**	**62.1**
KL + Student pixel GAN	**2514.1**	**1535.5**	**1072.8**	63.3	65.4

Figure 2: Comparison between teacher-based latent GAN and our student-based pixel GAN. Training efficiency is measured with batch size 8 per GPU. Lower is better for all efficiency metrics.

on image-only encoders such as DINO or SAM [3, 24, 15]. While these critics improve visual realism and structure, they provide no explicit feedback on text alignment. Conversely, vision-language models such as CLIP or SigLIP [26, 36] provide strong image–text matching signals, but their similarity scores do not directly model image realism or structural artifacts.

A natural solution is to combine image-only and vision-language VFMs. However, directly concatenating their features is suboptimal because they are trained with different objectives and live in different representation spaces: SigLIP is optimized for image–text contrastive alignment, whereas DINOv2 is trained with image-only self-supervision. Early feature fusion therefore entangles semantic and structural cues, making the adversarial signal harder to optimize.

Decoupled dual-branch VFM Discriminator. We propose a decoupled dual-branch VFM discriminator. The first branch is a **conditional semantic critic** based on frozen SigLIP, which evaluates prompt–image compatibility and provides direct adversarial supervision for text alignment:

$$\ell_{\text{SigLIP}}(x, y) = \phi_{\text{SigLIP}}(x, y), \tag{9}$$

where ϕ_{SigLIP} denotes the frozen SigLIP image–text scoring function. Since we use the native logits rather than re-projecting or mixing its embeddings with other VFM features, the semantic branch preserves SigLIP's pretrained vision-language prior.

The second branch is an **unconditional structural critic** based on frozen DINOv2. We aggregate its image tokens into a global descriptor and map it to a scalar realism logit with a trainable MLP head:

$$\ell_{\text{DINO}}(x) = g_{\text{structure}}\left(\mathcal{F}_{\text{DINO}}(x)\right), \qquad \mathcal{F}_{\text{DINO}}(x) = P\left(\phi_{\text{DINO}}(x)\right), \tag{10}$$

where ϕ_{DINO} is the frozen DINOv2 backbone and P denotes token-wise averaging. Since this branch receives no text conditioning, it focuses on image realism, global structure, and visual coherence.

Addressing Feature Space Misalignment. After the semantic and structural branches produce their own scalar logits, we combine them using a lightweight trainable fusion MLP g_{fuse}:

$$D_{\text{VFM}}(x, y) = g_{\text{fuse}}\left([\ell_{\text{SigLIP}}(x, y); \ell_{\text{DINO}}(x)]\right). \tag{11}$$

Crucially, we fuse only scalar logits rather than intermediate embeddings, preserving each VFM's native role: SigLIP captures prompt–image alignment, while DINOv2 captures realism and structure. This avoids the feature-space misalignment of early fusion, while also avoiding the manual loss balancing required when using separate SigLIP and DINOv2 adversarial discriminators.

VFM adversarial objective. Let y be a text prompt, $x_{\text{ref}} \sim p_{\text{data}}$ be the corresponding real image, and $x_0 = G_\theta(\xi, y)$ be a one-step generated image with $\xi \sim \mathcal{N}(0, I)$. The VFM discriminator is trained with the standard non-saturating adversarial objective:

$$\mathcal{L}_{GAN}^{\text{VFM}} = \mathbb{E}_{x_{\text{ref}} \sim p_{\text{data}},\, y}\left[f\left(-D_{\text{VFM}}(x_{\text{ref}}, y)\right)\right] + \mathbb{E}_{\xi \sim N(0,I), y, x_0 = G_\theta(\xi, y)}\left[f\left(-D_{\text{VFM}}(x_0, y)\right)\right]. \tag{12}$$

where $f(t) = -\log(1 + e^{-t})$ denotes the Softplus function. All VFM backbones are frozen. We optimize only the DINOv2 structural head g_{str} and the fusion head g_{fuse}.

3.4 Spectral Curriculum Objectives

A key bottleneck in heterogeneous distillation is the spectral mismatch between VAE-projected teacher latents and RGB images. Although the encoder $\mathcal{E}$ provides a differentiable bridge, modern VAEs can amplify high-frequency components [31] in latent space. However, these components are distinct from the meaningful textures discussed in Sec. 3.2. Instead, they may reflect non-semantic latent fluctuations that are normally absorbed or reinterpreted by VAE decoder $\mathcal{D}$.

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This becomes problematic for a VAE-free pixel student. Although the high-frequency components originate in the teacher's latent space, their gradients are backpropagate through $\mathcal{E}$ and directly shape the student's RGB output. However, the student has no decoder $\mathcal{D}$ to absorb them these high frequencies. The issue is severe in one-step generation: unlike multi-step diffusion, which builds images through a coarse-to-fine trajectory [41], the student must learn this process in a single forward pass. Without controlling the teacher's frequency contents, early optimization can over-emphasize unstable high-frequency components before learning coherent low-frequency structure, resulting in grid-like or checkerboard RGB artifacts, as illustrated in Fig. 5b.

We propose *Spectral Curriculum Distillation*, which controls frequency contributions in both pixel and score-update spaces. It imposes an explicit ceiling on high-frequency weights to prevent amplified latent frequencies from dominating supervision. In pixel space, a scheduler-weighted band-profile loss regularizes the image spectrum. In score space, we reweight the DMD update in the frequency domain before backpropagation. Together, these objectives encourage stable low-frequency structure early in training, then gradually introduce controlled high-frequency details.

Frequency band profile. Let $\mathcal{F}(x)$ denote the 2D FFT of image x, and let $\{B_k\}_{k=1}^{K}$ partition the Fourier domain into radial bands from low to high frequency. We define the relative spectral profile of each band $p_k(x)$ as follows:

$$p_k(x) = \frac{e_k(x)}{\sum_{j=1}^{K} e_j(x) + \varepsilon}, \qquad \text{where } e_k(x) = \frac{1}{C|B_k|} \sum_{c=1}^{C} \sum_{\omega \in B_k} \left| \mathcal{F}(x_c)(\omega) \right|. \tag{13}$$

Here, $e_k(x)$ is the average Fourier amplitude in band B_k across total C channels, x_c is the image at the c-th channel, and ε prevents division by zero. By normalizing across bands, $p_k(x)$ discards phase and absolute scale, retaining only the relative distribution of spectral energy.

Curriculum over spectral bands. Let τ denote the training step and N the curriculum length. We define the curriculum progress $\rho(\tau)$ and the scheduled highest-frequency weight $h(\tau)$ as

$$\rho(\tau) = \min\left(\frac{\tau}{N}, 1\right), \qquad h(\tau) = w_{\text{high}}^{(0)} + \rho(\tau)\left(w_{\text{high}}^{(1)} - w_{\text{high}}^{(0)}\right), \tag{14}$$

where $w_{\text{high}}^{(0)}$ and $w_{\text{high}}^{(1)}$ denote the initial and final weights assigned to the highest-frequency band. Thus, $h(\tau)$ increases linearly from $w_{\text{high}}^{(0)}$ to $w_{\text{high}}^{(1)}$ over the first N training steps and remains fixed afterward. For each band k, we define its normalized radial position as $\gamma_k = (k-1)/(K-1)$, where $\gamma_k = 0$ and $\gamma_k = 1$ correspond to the lowest- and highest-frequency bands, respectively. The band weight is linearly interpolated between w_{low} and the scheduled high-frequency weight $h(\tau)$, then normalized by the mean weight across all K bands:

$$\bar{\omega}_k(\tau) = w_{\text{low}} + (h(\tau) - w_{\text{low}})\,\gamma_k, \qquad \omega_k(\tau) = \frac{\bar{\omega}_k(\tau)}{\frac{1}{K}\sum_{j=1}^{K} \bar{\omega}_j(\tau)}. \tag{15}$$

Here, w_{low} is the fixed lowest-band weight. Normalization keeps the average band weight equal to one, preserving the overall loss scale during training. By default, we set $w_{\text{high}}^{(0)} < w_{\text{low}}$ and $w_{\text{high}}^{(1)} \approx w_{\text{low}}$, so high frequencies are suppressed early and gradually restored to balanced supervision.

Spectrum-profile loss. Since image quality is ultimately measured in RGB space, we regularize the generated image x_0 using a pixel-space spectral reference. However, the teacher operates in latent space, so we estimate clean latent $\hat{z}_{0,\psi}(z_t, t, y)$ using clean prediction formula in Eq. 3, and decode it with the frozen VAE decoder $\mathcal{D}$. The decoded image serves only as a teacher-implied spectral target, not as a pixel-level reconstruction target. This lets the student follow the teacher's coarse spectral structure while still preserving freedom in phase, texture, and exact RGB appearance:

$$x_{\text{teacher}} = \mathcal{D}\left(\hat{z}_{0,\psi}(z_t, t, y)\right). \tag{16}$$

Given the student image x_s and the teacher-implied image x_{teacher}, band-profile loss at step τ is:

$$\mathcal{L}_{\text{spec}} = \frac{1}{K}\sum_{k=1}^{K} \omega_k(\tau)\left(\log p_k(x_0) - \log p_k(x_{\text{teacher}})\right)^2. \tag{17}$$

Since $p_k(\cdot)$ is normalized, $\mathcal{L}_{\text{spec}}$ constrains the distribution of spectral energy rather than the absolute Fourier amplitudes. Applying the loss in log space further makes each term measure a overall relative

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mismatch. Thus, the loss penalizes relative over- or under-representation of each frequency band, rather than enforcing pixel-level reconstruction or matching absolute spectral energy.

Spectrum-weighted Score Gradient. While $\mathcal{L}_{\text{spec}}$ regularizes the spectrum of the final RGB output, the primary knowledge transfer relies on the score objective $\mathcal{L}_{\text{DMD}}$. It is therefore important to impose the same spectral curriculum directly on the score update, rather than only correcting the generated image after the update has been applied. Given a score map s at training step τ, we apply the spectral curriculum by transforming it into the Fourier domain, multiplying each band by its curriculum weight, and mapping it back to the spatial domain for spectrum-adjusted score:

$$\mathcal{W}(s) = \mathcal{F}^{-1}\left(\left[\sum_{k=1}^{K} \omega_k(\tau)\mathcal{M}_{B_k}\right] \odot \mathcal{F}(s)\right), \tag{18}$$

where $\mathcal{M}_{B_k}$ is the binary mask for frequency band B_k and $\mathcal{F}^{-1}$ denotes the inverse FFT function. The spectrum-adjusted scores are $s^{\text{adj}}_{\text{real}} = \mathcal{W}(s_{\text{real}})$ and $s^{\text{adj}}_{\text{fake}} = \mathcal{W}(s_{\text{fake}})$, and the spectrum-adjusted KL gradient becomes:

$$\nabla_\theta \mathcal{L}^{spec}_{\text{KL}} = -\mathbb{E}_{z,t,z_t}\left[\left(s^{\text{adj}}_{\text{real}}(z_t,t,y) - s^{\text{adj}}_{\text{fake}}(z_t,t,y)\right)\frac{dG_\theta(z_T,y)}{d\theta}\right]. \tag{19}$$

3.5 Stage-I: Full-Objective Distillation

In the first stage, we train a full-objective checkpoint using the spectrum-weighted score KL objective together with our pixel-space adversarial, VFM adversarial, and spectrum objectives:

$$\mathcal{L}_{\text{Stage-I}} = \lambda_{\text{KL}}\mathcal{L}^{\text{spec}}_{\text{KL}} + \lambda^{\text{Student}}_{\text{GAN}}\mathcal{L}^{\text{Student}}_{\text{GAN}} + \lambda^{\text{VFM}}_{\text{GAN}}\mathcal{L}^{\text{VFM}}_{\text{GAN}} + \lambda_{\text{spec}}\mathcal{L}_{\text{spec}}. \tag{20}$$

Here, λ_{KL}, $\lambda^{\text{Student}}_{\text{GAN}}$, $\lambda^{\text{VFM}}_{\text{GAN}}$, and λ_{spec} control the weights of the spectrum-weighted KL, student adversarial, VFM adversarial, and spectrum-profile objectives, respectively. This stage produces the full-objective checkpoint used for preference-calibrated model merging in Sec. 3.6.

3.6 Preference-Calibrated Model Merging

While the full-objective checkpoint improves RGB fidelity, semantic guidance, and spectral stability, it does not explicitly optimize human preference. However, naively adding a preference loss to Eq. 20 can improve aesthetics but weaken compositional or fine-grained grounding and complicate multi-objective optimization.

Following [2, 8], we instead use model merging to combine complementary checkpoints. We train a separate preference-tuned checkpoint using the KL loss in Eq. 5 and a bounded HPSv2 [38] objective. Let $R_{\text{HPS}}(x_0, y)$ be the frozen HPSv2 reward for $x_0 = G_\theta(\xi, y)$. We use the clamped objective:

$$\mathcal{L}_{\text{HPS}} = \max\left(0,\, \tau_{\text{HPS}} - R_{\text{HPS}}(x_0, y)\right), \tag{21}$$

Figure 3: Effects of τ_{HPS} in the preference-tuned checkpoint.

where τ_{HPS} is a fixed target reward. The loss is active only below this threshold, preventing unbounded reward optimization. As shown in Fig. 3, increasing τ_{HPS} initially improves fidelity and fine details, but overly large values lead to reward-hacking artifacts such as oversharpening, excessive saturation, and overly busy backgrounds. We therefore use $\tau_{\text{HPS}} = 0.35$, which provides a good trade-off between preference improvement and visual stability. Let θ_{full} denote the full-objective checkpoint in Sec. 3.5, and θ_{pref} the preference-tuned checkpoint. The final model is obtained by linear interpolation:

$$\theta_{\text{final}} = (1-\alpha)\theta_{\text{full}} + \alpha\theta_{\text{pref}}, \tag{22}$$

where α controls the preference-calibration strength. This preserves the grounding and fidelity of the full-objective model while incorporating the aesthetic gains of preference tuning.

3.7 Qualitative Results

Fig. 4 compares our enhanced method with naive DMD2 under the same latent-teacher and pixel-student pairs. Across diverse prompts, our enhanced heterogeneous distillation produces sharper

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Table 1: Quantitative comparison of teacher base models and one-step distilled students across pixel and latent spaces. Higher is better for evaluation metrics (↑). MPS is reported as the average of the two sub-scores. Our method is highlighted in green. Best results within each block are in **bold**.

Teacher	Method	Student	NFEs	Size	Throughput (img/s)	HPSv2 ↑	HPSv3 ↑	GenEval ↑	DPG ↑	IR ↑	MPS ↑
Teacher Base Models											
Flux.2			50	3.8B	0.09	28.11	10.30	75.79	0.76	0.81	13.52
SD 3.5			50	2.5B	0.25	29.52	10.70	67.53	0.84	0.87	13.86
PixelGen			50	1.1B	0.50	26.01	7.95	73.96	0.73	0.26	12.62
PixNerd			50	1.2B	0.46	27.76	9.93	70.01	0.76	0.53	12.56
Pixel-Space Distillation											
		Student: PixelGen	*NFEs: 1, Size: 1.1B, Throughput: 24.42 img/s*								
PixelGen	DMD2					26.01	7.46	67.86	0.75	0.35	12.71
	Ours					**29.68** (+3.67)	**9.83** (+2.37)	**70.31** (+2.45)	**0.79** (+0.04)	**0.60** (+0.25)	**13.61** (+0.90)
Flux.2	DMD2					27.86	9.41	66.26	0.75	0.38	12.68
	Ours					**32.95** (+5.09)	**11.03** (+1.62)	**73.72** (+7.46)	**0.78** (+0.03)	**0.81** (+0.43)	**14.66** (+1.98)
SD 3.5	DMD2					28.15	9.49	72.80	0.80	0.60	13.14
	Ours					**33.47** (+5.32)	**10.96** (+1.47)	**75.76** (+2.96)	**0.82** (+0.02)	**0.80** (+0.20)	**14.06** (+0.92)
		Student: PixNerd	*NFEs: 1, Size: 1.2B, Throughput: 22.35 img/s*								
PixNerd	DMD2					25.67	6.33	58.42	0.67	0.30	11.17
	Ours					**31.88** (+6.21)	**10.81** (+4.48)	**68.81** (+10.39)	**0.79** (+0.12)	**1.03** (+0.73)	**13.95** (+2.78)
Flux.2	DMD2					28.27	9.50	67.82	0.77	0.51	12.63
	Ours					**33.44** (+5.17)	**13.05** (+3.55)	**75.48** (+7.66)	**0.82** (+0.05)	**0.97** (+0.46)	**14.51** (+1.88)
SD 3.5	DMD2					29.62	8.78	61.71	0.76	0.76	13.33
	Ours					**34.32** (+4.70)	**12.47** (+3.69)	**71.90** (+10.19)	**0.83** (+0.07)	**0.98** (+0.22)	**14.64** (+1.31)
Latent-Space Distillation			*Teachers and students share the same architecture.*								
Flux.2	DMD2		1	3.8B	3.96	27.80	9.61	72.84	0.91	0.53	13.57
SD 3.5	DMD2		1	2.5B	9.22	29.65	8.04	58.83	0.83	0.90	13.33
SD 1.5	DMD2		1	0.8B	16.29	26.71	6.03	0.48	0.56	0.59	11.62

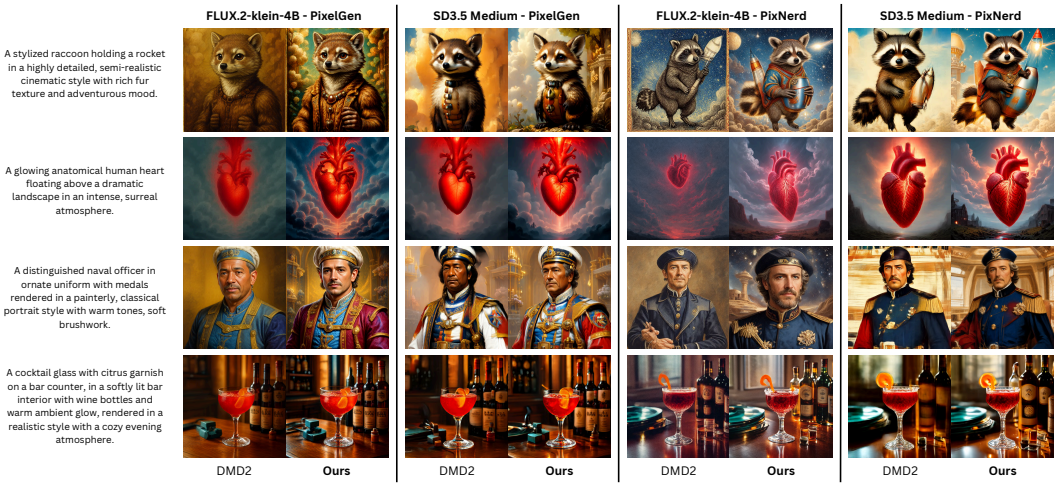

Figure 4: **Qualitative comparison of our enhanced heterogeneous score distillation.** For each teacher–student pair, we compare naive DMD2 with our method using the same prompt.

textures, cleaner object boundaries, and more faithful prompt details. These results visually support the quantitative gains in Tab. 2, showing that our framework improves both fine-grained fidelity and semantic grounding over the naive latent-to-pixel DMD2 baseline.

4 Experiments

4.1 Experimental Setup

Distillation Settings. First, we report the original multi-step teachers as reference baselines. Second, we evaluate same-space distillation, including pixel-to-pixel and latent-to-latent setups. Third, we study our target heterogeneous setting, where a strong latent teacher distills into a VAE-free pixel

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student. For each pixel student, we compare naive DMD2 distillation with our enhanced framework under the same teacher–student pair.

Student Models. For pixel-space students, we use pretrained multi-step pixel diffusion models, PixelGen [21] and PixNerd [37]. For latent-to-latent distillation baselines, we use Flux.2 Klein-4B [16] and SD 3.5 Medium [9] as latent students.

Teacher Models. For pixel-space students, we use both pixel teachers, PixelGen and PixNerd, and latent teachers, FLUX.2-Klein-4B and SD3.5-Medium. The latent teachers defines our heterogeneous latent-to-pixel setting. For latent-to-latent baselines, the teacher and student remain within the same model family, using either FLUX.2-Klein-4B or SD3.5-Medium.

Data and Implementation Details. We provide full training details, hyperparameter settings, and dataset construction in Appendix sections F and G.

Metrics. We use widely adopted metrics, including HPSv2 [38], HPSv3 [20], ImageReward [39], MPS [45], GenEval [12] and DPG [14]. HPSv2 and HPSv3 measure human preference alignment, respectively, while ImageReward provides a learned proxy for human judgments. We average MPS scores on both HPSv2 and ImageReward prompt sets to capture multi-dimensional quality. Finally, GenEval assesses compositional accuracy, while DPG evaluates fine-grained text–image grounding.

Throughput. We measure inference throughput at 512×512 resolution on a single NVIDIA H100 GPU, averaged over 100 generated samples. Throughput is reported in images per second, where higher indicates faster inference. For detailed ablations, please refer to Appendix section D.

4.2 Main Quantitative Results

Tab. 2 shows that our enhanced framework consistently outperforms naive DMD2 under the same teacher–student pairs, confirming that standard distillation under-constrains one-step pixel students. Across both PixelGen and PixNerd, latent teachers yield large gains in HPSv2, reaching up to **34.32**, with similar improvements across preference, compositional, and grounding metrics. Notably, strong LDM teachers produce better one-step pixel students than pixel-to-pixel distillation, supporting our claim that LDMs can effectively supervise VAE-free pixel generation. Our best model surpasses all one-step baselines and exceeds multi-step teachers on key preference metrics, while preserving the speed advantage of pixel generation: PixelGen and PixNerd run at **24.42** and **22.35** img/s, giving 2.4–6.2$\times$ speedups over latent baselines and about $\mathbf{1.5}\times$ over the smaller one-step SD1.5 baseline.

Some one-step students outperform their multi-step teachers on certain metrics, a phenomenon also reported in prior distillation methods [42, 4, 19, 1, 46]. This is expected because score-based distillation provides distributional guidance rather than paired targets: the student inherits the teacher's generative prior but is not forced to reproduce its exact sampling trajectory. Additional objectives can further calibrate the student toward preference, grounding, or visual-quality metrics, allowing it to excel on specific benchmarks. Thus, multi-step teachers are strong references but not strict upper bounds for every metric. We emphasize, however, that these gains indicate metric-specific improvements, not universal superiority over the teacher.

5 Conclusion

We present a heterogeneous distillation framework that transfers strong latent diffusion teachers into one-step VAE-free pixel students. By adding pixel adversarial supervision, decoupled VFM guidance, spectral curriculum objectives, and preference-calibrated merging, our method improves over naive DMD2 while preserving fast direct RGB inference. Our results show that pixel diffusion is a promising direction for efficient one-step text-to-image generation.

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Appendix

A Related Work

A.1 Latent and Pixel Diffusion Models

State-of-the-art text-to-image models are largely based on latent diffusion models [27, 25, 9, 16, 2], which benefit from scalable architectures and efficient denoising in a compressed latent space. While this design has enabled strong visual fidelity and prompt alignment, it also requires VAE decoding to reconstruct the final RGB image.Pixel-space text-to-image models remove this decoder by generating directly in RGB space. Early pixel-space models [17] lagged behind LDMs due to the cost of full-resolution RGB denoising, but recent methods [21, 37, 44] have improved scalability and quality, narrowing the gap with latent diffusion models. This renewed strength makes pixel diffusion especially attractive for one-step generation, where removing the VAE decoder can directly reduce inference cost.

A.2 Diffusion Distillation

Diffusion distillation compresses a multi-step sampler into a one- or few-step generator while preserving the teacher's generation quality. Early progressive distillation methods [28] shorten the sampling trajectory through multiple stages while maintaining the original model's behavior. Consistency-based methods [34, 33, 7, 18, 11, 10, 22] train models to map noisy samples at different timesteps to a shared trajectory endpoint. Other approaches add adversarial supervision: ADD [30] and LADD [29] add adversarial perceptual guidance to improve sharpness and realism. Meanwhile, score-based distillation methods [23, 8, 43, 42, 4, 19] use teacher-derived scores to guide the student toward the image distribution learned by the teacher. Hybrid methods [6] combine multiple objectives, such as trajectory-level and score-based objectives.Among these approaches, score-based distillation achieves the most competitive performance in one-step regimes. In this work, we instantiate our heterogeneous distillation framework on top of DMD [43, 42] due to its strong empirical performance and broad adoption. However, our components address the latent-teacher/pixel-student gap and are not specifically tied to DMD; they can in principle be combined with other score-based distillation methods.

A.3 Fast Fourier Transform

Fourier transform analyzes signals in the frequency domain by decomposing them into sinusoidal components. In practice, we operate on discrete data and employ the Discrete Fourier Transform (DFT), which maps a finite signal $x \in \mathbb{R}^N$ to its complex-valued frequency representation $X = \mathcal{F}(x)$:

$$X_k = \sum_{n=0}^{N-1} x_n \, e^{-i2\pi kn/N}, \quad k = 0, \ldots, N-1. \tag{23}$$

The Fast Fourier Transform (FFT) computes this transformation efficiently in $\mathcal{O}(N \log N)$ time using divide-and-conquer algorithms such as Cooley-Tukey, significantly improving over the $\mathcal{O}(N^2)$ complexity of a naive DFT. For images, 2D FFT decomposes spatial signals into frequency components, where low frequencies capture global structure and high frequencies encode fine-grained details.

B Full-Objective Distillation vs. Merged Models

Tab section B compares the pre-merged full-objective checkpoint with the final preference-merged model. The full-objective checkpoint already improves over DMD2, especially on GenEval and DPG, showing the benefit of pixel-space, VFM, and spectral supervision. Preference-calibrated merging further improves HPSv2, IR, and MPS in most settings, while preserving most of the compositional and grounding gains. In some cases, the pre-merged model has slightly higher GenEval or DPG, indicating a trade-off between preference alignment and fine-grained grounding. Overall, merging provides a simple way to recover stronger human-preference scores without collapsing the gains from the full heterogeneous objective.

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Table 2: Quantitative comparison between pre-merged full-objective checkpoints and final preference-merged models. † denotes the pre-merged full-objective checkpoint before preference-calibrated merging. Higher is better for all metrics (↑). Flux.2 denotes FLUX.2-Klein-4B, and SD 3.5 denotes Stable Diffusion 3.5 Medium.

Teacher	Method	HPSv2 ↑	HPSv3 ↑	GenEval ↑	DPG ↑	IR ↑	MPS ↑
Pixel-Space Distillation							
Student: PixelGen *NFEs: 1, Size: 1.1B, Throughput: 24.42 img/s*							
PixelGen	DMD2	26.01	7.46	67.86	0.75	0.35	12.71
	Ours†	**28.01** (+2.00)	**8.89** (+1.43)	**70.75** (+2.89)	**0.79** (+0.04)	**0.49** (+0.14)	**13.17** (+0.46)
	Ours	**29.68** (+3.67)	**9.83** (+2.37)	**70.31** (+2.45)	**0.79** (+0.04)	**0.60** (+0.25)	**13.61** (+0.90)
Flux.2	DMD2	27.86	9.41	66.26	0.75	0.38	12.68
	Ours†	**29.97** (+2.11)	**10.89** (+1.48)	**74.25** (+7.99)	**0.79** (+0.04)	**0.61** (+0.23)	**13.92** (+1.24)
	Ours	**32.95** (+5.09)	**11.03** (+1.62)	**73.72** (+7.46)	**0.78** (+0.03)	**0.81** (+0.43)	**14.66** (+1.98)
SD 3.5	DMD2	28.15	9.49	72.80	0.80	0.60	13.14
	Ours†	**31.29** (+3.14)	**10.05** (+0.56)	**76.52** (+3.72)	**0.84** (+0.04)	**0.72** (+0.12)	**14.00** (+0.86)
	Ours	**33.47** (+5.32)	**10.96** (+1.47)	**75.76** (+2.96)	**0.82** (+0.02)	**0.80** (+0.20)	**14.06** (+0.92)
Student: PixNerd *NFEs: 1, Size: 1.2B, Throughput: 22.35 img/s*							
PixNerd	DMD2	25.67	6.33	58.42	0.67	0.30	11.17
	Ours†	**29.14** (+3.47)	**8.85** (+2.52)	**70.27** (+11.85)	**0.80** (+0.13)	**0.76** (+0.46)	**12.20** (+1.03)
	Ours	**31.88** (+6.21)	**10.81** (+4.48)	**68.81** (+10.39)	**0.79** (+0.12)	**1.03** (+0.73)	**13.95** (+2.78)
Flux.2	DMD2	28.27	9.50	67.82	0.77	0.51	12.63
	Ours†	**29.48** (+1.21)	**13.57** (+4.07)	**74.00** (+6.18)	**0.86** (+0.09)	**0.79** (+0.28)	**13.52** (+0.89)
	Ours	**33.44** (+5.17)	**13.05** (+3.55)	**75.48** (+7.66)	**0.82** (+0.05)	**0.97** (+0.46)	**14.51** (+1.88)
SD 3.5	DMD2	29.62	8.78	61.71	0.76	0.76	13.33
	Ours†	**32.06** (+2.44)	**10.82** (+2.04)	**72.64** (+10.93)	**0.85** (+0.09)	**0.88** (+0.12)	**14.24** (+0.91)
	Ours	**34.32** (+4.70)	**12.47** (+3.69)	**71.90** (+10.19)	**0.83** (+0.07)	**0.98** (+0.22)	**14.64** (+1.31)

Table 3: Ablation studies of the proposed design choices for. Unless otherwise specified, all ablations use FLUX.2-Klein as the latent teacher and PixelGen as student. We evaluate the contribution of each major component and the effect of each loss within the spectral objectives. Higher is better for all metrics (↑). Selected variants are in green, and best results within each subtable are in **bold**.

(a) Top-up ablation. Second-best are underlined.

Method	HPSv2 ↑	HPSv3 ↑	GenEval ↑	DPG ↑	IR ↑	MPS ↑
DMD2 ($\mathcal{L}_{\text{KL}} + \mathcal{L}_{\text{GAN}}^{DMD2}$)	27.86	9.41	66.26	0.75	0.38	12.68
Enhanced Heterogeneous Score Distillation						
[A1]: $\mathcal{L}_{\text{KL}} + \mathcal{L}_{\text{GAN}}^{Student}$ (Sec. 3.2)	28.01	9.83	67.73	0.75	0.41	12.94
[A2]: **[A1]** $+\mathcal{L}_{\text{GAN}}^{VFM}$ (Sec. 3.3)	28.91	10.25	70.82	0.78	0.45	13.23
[A3]: **[A2]** $+\mathcal{L}_{\text{spec}}$ (Sec. 3.4)	29.97	10.89	**74.25**	**0.79**	0.61	13.92
[A4]: $\mathcal{L}_{\text{KL}} + \mathcal{L}_{\text{HPS}}$ (Sec. 3.6)	**33.62**	**11.72**	69.15	0.72	**0.83**	**14.92**
Merge [A3] & [A4]	32.95	11.03	73.72	0.78	0.81	14.66

(b) Spectral curriculum ablation.

Method	HPSv2 ↑	HPSv3 ↑	GenEval ↑	DPG ↑	IR ↑	MPS ↑
Baseline: $\mathcal{L}_{\text{KL}} + \mathcal{L}_{\text{GAN}}^{Student} + \mathcal{L}_{\text{GAN}}^{VFM}$	28.91	10.25	70.82	0.78	0.45	13.23
[A1]: Baseline + $\mathcal{L}_{\text{spec}}$ (No $\omega_k(\tau)$)	29.26	10.42	72.51	0.79	0.47	13.40
[A2]: Baseline + Replace $\mathcal{L}_{\text{KL}}$ w/ $\mathcal{L}_{\text{KL}}^{\text{spec}}$	29.30	10.49	71.94	0.78	0.50	13.38
[A3]: [A1] + [A2] (No $\omega_k(\tau)$)	29.58	10.63	73.65	**0.80**	**0.52**	13.57
[A4]: [A3] using $\mathcal{L}_{\text{KL}}^{\text{spec}}$ w/ $\omega_k(\tau)$	**29.97**	**10.89**	**73.72**	0.79	0.51	**13.92**

C Ablation Study

C.1 Top-up ablation

In Tab. 3a, we ablate the contribution of each proposed component improves over DMD2. Student-based pixel GAN gives marginal but consistent gains while maintaining low training latency. Both the decoupled VFM GAN and spectral objectives improves prompt alignment and fine-grained grounding, reflected by GenEval and DPG. Reward tuning alone achieves the best preference scores but substantially degrades other metrics. Merging the reward-tuned checkpoint with the full-objective checkpoint preserves most preference gains while preserving GenEval and DPG, yielding the best overall trade-off.

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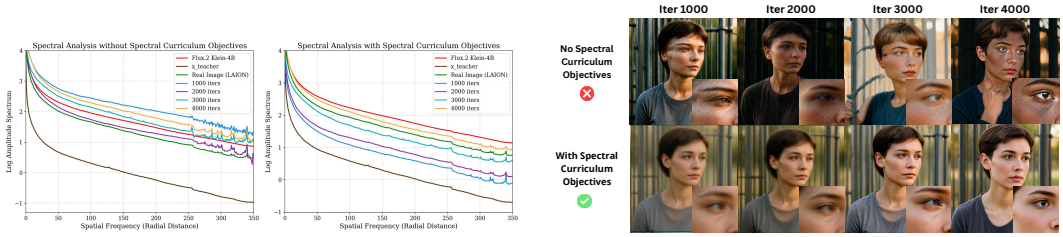

(a) Frequency-pattern evolution.

(b) Qualitative training dynamics.

Figure 5: Effect of the spectral curriculum during heterogeneous pixel distillation. Left: we compute the 2D FFT of generated RGB images across training iterations, radially average Fourier amplitudes, and compare the log-amplitude spectra against real LAION images, the 50-step Flux.2-Klein-4B teacher, and teacher-implied images x_{teacher}. Right: Generated samples and zoomed-in eye crops across iterations. Without the spectral curriculum, training exhibits irregular checkerboard artifacts, with no clear coarse-to-fine progression. With the spectral curriculum, frequency evolution is smoother and later iterations recover sharper details with fewer checkerboard artifacts.

C.2 Spectral Curriculum Objectives

Top-up ablation. In Tab. 3b, we ablate the pixel-space loss $\mathcal{L}_{\text{spec}}$, the score-space objective $\mathcal{L}_{\text{KL}}^{\text{spec}}$, and the curriculum weights $\omega_k(\tau)$. Starting from the KL + pixel GAN + VFM GAN baseline, both spectrum losses improve most metrics, with $\mathcal{L}_{\text{spec}}$ giving larger gains on GenEval/DPG and $\mathcal{L}_{\text{KL}}^{\text{spec}}$ giving stronger preference/reward gains. Combining them further improves performance, and adding $\omega_k(\tau)$ gives the best HPSv2, HPSv3, GenEval, and MPS while keeping DPG and IR comparable.

Frequency Patterns during training. In Fig. 5a, we visualize the student's frequency evolution. Without the spectral curriculum, the student exhibits large high-frequency fluctuations, sometimes exceeding the Flux.2 teacher, suggesting that latent score gradients inject amplified frequency components directly into RGB space. Since the pixel student has no VAE decoder to absorb these signals, they can manifest as unstable artifacts, as in Fig. 5b. With the spectral curriculum, the high-frequency tail becomes smoother and the iteration curves evolve more orderly: early iterations suppress high frequencies, while later iterations gradually approach the real-image and teacher spectra. This is consistent with a controlled coarse-to-fine learning process.

C.3 VFM Adversarial Objectives

Model selection. Tab. 4b ablates the VFM backbones. SigLIP-0.4B improves prompt alignment, while DINOv2 gives complementary visual-quality gains. Their combination achieves the best grounding trade-off, improving GenEval from 67.73 to 70.82 and DPG from 0.75 to 0.78. Qwen3.5-0.8B+DINOv2 slightly improves preference scores, but the gains are marginal and require roughly **3**$\times$ longer training. We therefore use SigLIP-0.4B+DINOv2 by default.

Branch Fusion. We compare three ways to fuse the conditional SigLIP branch and the image-only DINOv2 branch. *Concat embeddings + MLP* projects both VFM embeddings to the same dimension, concatenates them, and predicts the discriminator logit with a single MLP. *Decoupled embedding fusion* maps SigLIP and DINOv2 embeddings with separate MLPs to compute logits, then concatenates them and predicts the final logit with another MLP. *Decoupled logit fusion* follow the same setup in Sec. 3.3. As shown in Tab. 4c, early embedding fusion performs worse than the baseline, suggesting that directly mixing SigLIP and DINOv2 features entangles incompatible representation spaces. Decoupled embedding fusion improves over early fusion, but remains weaker than logit fusion. Our proposed decoupled logit fusion performs best overall.

D Inference Efficiency

We further analyze where the inference speedup comes from. All measurements are conducted at 512×512 resolution on a single NVIDIA H100 GPU and averaged over 100 generated samples. For latent-space models, we report one-step throughput both with and without VAE decoding, and

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Table 4: Additional VFM ablations. Unless otherwise specified, all ablations use FLUX.2-Klein as teacher and PixelGen as student. Higher is better for all metrics (↑). Selected variants are highlighted in green.

(b) VFM selection. Second-best are underlined.

Method	HPSv2 ↑	HPSv3 ↑	GenEval ↑	DPG ↑	IR ↑	MPS ↑
Baseline: $\mathcal{L}_{\text{KL}} + \mathcal{L}_{\text{GAN}}^{Student}$	28.01	9.83	67.73	0.75	0.41	12.94
+ SigLIP 0.4B	28.58	10.01	<u>70.10</u>	**0.78**	0.43	13.17
+ DINOv2	28.62	10.03	68.03	0.76	0.42	13.07
+ SigLIP 0.4B, SAM	28.59	9.98	68.26	0.72	0.40	12.90
+ Qwen 3.5 0.8B, DINOv2	**29.01**	**10.32**	69.93	<u>0.77</u>	**0.47**	**13.31**
+ SigLIP 0.4B, DINOv2	<u>28.91</u>	<u>10.25</u>	**70.82**	**0.78**	<u>0.45</u>	<u>13.23</u>

(c) VFM branch fusion.

Method	HPSv2 ↑	HPSv3 ↑	GenEval ↑	DPG ↑	IR ↑	MPS ↑
Baseline: $\mathcal{L}_{\text{KL}} + \mathcal{L}_{\text{GAN}}^{Student}$	28.01	9.83	67.73	0.75	0.41	12.94
Concat embedding + MLP	27.94	9.10	65.01	0.73	0.36	12.67
Decoupled embedding fusion	28.14	9.96	68.38	**0.78**	0.42	12.88
Decoupled logit fusion	**28.91**	**10.25**	**70.82**	**0.78**	**0.45**	**13.23**

compute the decoder overhead from latency as

$$100 \times \left(\frac{\text{Latency}_{\text{w/VAE}}}{\text{Latency}_{\text{w/oVAE}}} - 1 \right).$$

For pixel-space models, generation directly produces RGB images, so there is no separate VAE decoding stage.

As shown in Tab. 5, VAE decoding becomes a visible bottleneck once denoising is reduced to one step. For SD1.5, the decoder more than doubles the FP32 one-step latency, giving a $+104.9\%$ overhead. For larger latent models, the relative overhead depends on the denoising backbone, but remains non-negligible: $+39.5\%$ for SD 3.5 and $+14.2\%$ for FLUX.2 in FP32, and $+57.1\%$ and $+75.2\%$ in BF16.

Pixel diffusion avoids this cost entirely because the model directly outputs RGB images. In FP32, PixelGen and PixNerd reach 24.31 and 22.35 img/s, compared with 16.93 img/s for SD1.5, 9.21 img/s for SD 3.5, and 3.92 img/s for FLUX.2 with VAE decoding. The advantage becomes larger in BF16, where PixelGen and PixNerd reach 48.09 and 44.27 img/s. This supports the central motivation of our work: after the denoising trajectory is collapsed to one NFE, removing the decoder becomes a direct source of end-to-end inference speedup.

E User Study

To evaluate perceptual quality and prompt alignment, we conducted a user study comparing our distilled one-step models against the DMD2 baseline.

Procedure. We collected responses from 300 participants on 100 side-by-side comparison cases. The cases were evenly split across four teacher–student configurations: FLUX.2–PixelGen, SD3.5–PixelGen, FLUX.2–PixNerd, and SD3.5–PixNerd, with 25 cases per configuration. For each case, participants were shown a text prompt and two anonymized images generated by our method and DMD2 under the same teacher–student setting. They selected the better image under two criteria:

- **Overall Image Quality:** This criterion reflects the overall fidelity, clarity, sharpness, and the absence of noticeable artifacts or unnatural details.
- **Textual Alignment:** This criterion measures consistency with the prompt content, objects, style, and scene description.

Results. As shown in Fig. 6, our method is preferred over DMD2 across all teacher–student configurations. For image quality, our method wins 53.0–62.0% of comparisons, with the strongest gains under FLUX.2 teachers for both PixelGen and PixNerd. This supports that bypassing latent-space adversarial supervision and using a student-side pixel discriminator improves texture sharpness, object boundaries, and visual fidelity. For textual alignment, our method also maintains a consistent advantage, winning 46.0–52.0% of comparisons despite higher tie rates. These results suggest that the decoupled VFM discriminator improves semantic grounding without sacrificing realism. Overall, the user study confirms that our heterogeneous distillation framework improves both perceived image quality and prompt alignment over naive DMD2.

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Table 5: Inference throughput and VAE overhead comparison at 512×512 resolution on H100. Throughput is measured in images per second. For latent-space models, VAE overhead is computed from one-step latency as $100 \times (\text{Latency}_{\text{w/VAE}}/\text{Latency}_{\text{w/oVAE}} - 1)$. Higher is better for throughput (↑), and lower is better for VAE overhead (↓).

Model	Precision	50-step Throughput ↑ (img/s)	1-step w/ VAE ↑ (img/s)	1-step w/o VAE ↑ (img/s)	VAE Overhead ↓ (%)
FP32					
Pixel-Space Models		*Direct RGB generation without VAE decoding*			
PixelGen	FP32	**0.50**	**24.31**	*direct RGB*	*VAE-free*
PixNerd	FP32	0.46	22.35	*direct RGB*	*VAE-free*
Latent-Space Models		*One-step VAE decoding overhead*			
SD1.5	FP32	*N/M*	16.93	34.68	**+104.9%**
SD 3.5	FP32	0.25	9.21	12.85	**+39.5%**
FLUX.2	FP32	0.09	3.92	4.47	**+14.2%**
BF16					
Pixel-Space Models		*Direct RGB generation without VAE decoding*			
PixelGen	BF16	**0.94**	**48.09**	*direct RGB*	*VAE-free*
PixNerd	BF16	0.91	44.27	*direct RGB*	*VAE-free*
Latent-Space Models		*One-step VAE decoding overhead*			
SD1.5	BF16	0.76	21.12	40.87	**+93.5%**
SD 3.5	BF16	0.47	15.06	23.67	**+57.1%**
FLUX.2	BF16	0.63	18.31	32.08	**+75.2%**

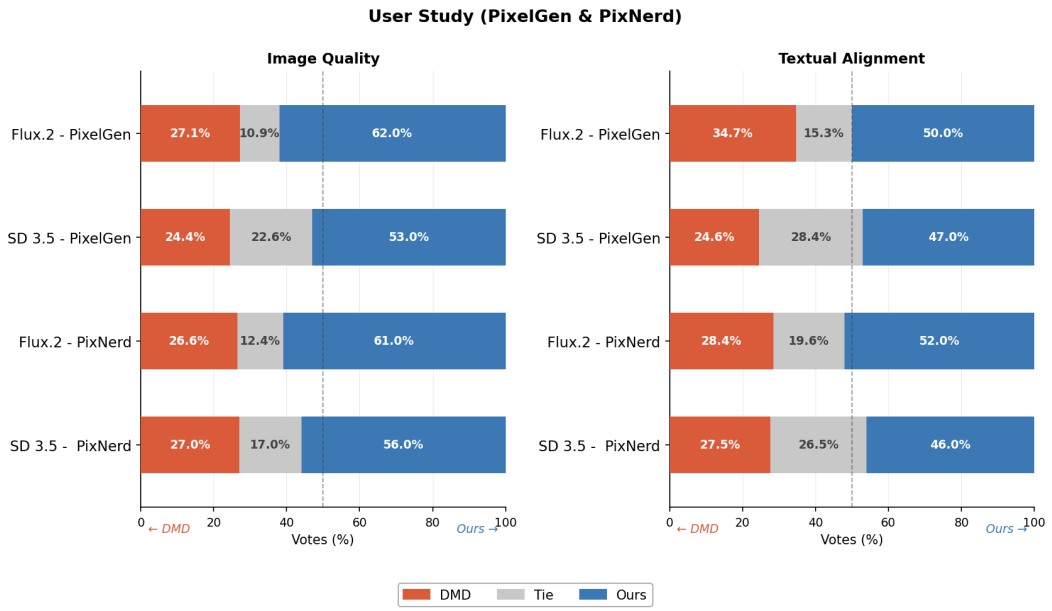

Figure 6: User study comparing our method against DMD2 across four teacher–student configurations. Bars show the percentage of votes for DMD2, tie, and ours under two criteria: image quality and textual alignment. Our method is consistently preferred for image quality and shows a clear advantage in textual alignment, validating the performance of our heterogeneous distillation framework.

F Training Data

Our training pipeline utilizes two specific data configurations:

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Table 6: Training hyperparameters for our heterogeneous one-step pixel distillation. Unless otherwise noted, optimization settings are shared across the latent-to-pixel configurations. Pixel students generate directly in RGB space and do not require VAE decoding at inference.

Category	Hyperparameter	Flux.2 → PixelGen	SD 3.5 → PixelGen	Flux.2 → PixNerd	SD 3.5 → PixNerd
Models	Teacher model	Flux.2-Klein-4B	SD 3.5 Medium	Flux.2-Klein-4B	SD 3.5 Medium
	Teacher size	3.8B	2.5B	3.8B	2.5B
	Teacher space	Latent	Latent	Latent	Latent
	Student model	PixelGen	PixelGen	PixNerd	PixNerd
	Student size	1.1B	1.1B	1.2B	1.2B
	Student space	RGB pixel	RGB pixel	RGB pixel	RGB pixel
	Student NFEs	1			
Optimization	Resolution	512×512			
	Hardware	8× NVIDIA H100 (80GB)			
	Precision	bf16			
	Batch size / GPU	8			
	Global batch size	64			
	Optimizer	AdamW (student: β_1=0.9, β_2=0.999, ϵ=10^{-8}, wd = 10^{-4}; fake score/LoRA: AdamW, wd = 0)			
	Student learning rate	1×10^{-6}			
	Fake score estimator learning rate	5×10^{-4}			
	Training steps	6,000			
	Gradient accumulation	1			
Discriminators	Student pixel-GAN learning rate	4×10^{-7}			
	Student pixel-GAN update ratio	1:1			
	VFM discriminator learning rate	4×10^{-7}			
	VFM discriminator update ratio	1:1			
	VFM semantic branch	google/siglip-so400m-patch14-384			
	VFM structural branch	DINOv2			
	VFM fusion	Decoupled logit fusion			
	Frozen VFM backbones	Yes			
Loss weights	λ_{KL}	0.1			
	$\lambda_{\mathrm{GAN}}^{\mathrm{Student}}$	0.1			
	$\lambda_{\mathrm{GAN}}^{\mathrm{VFM}}$	0.01			
	λ_{spec}	0.02			
	λ_{HPS}	0.005			
	HPS threshold τ_{HPS}	0.35			
	HPS reward model	HPSv2			
Spectrum curriculum	Number of frequency bands K	8			
	Curriculum length N	4,000			
	Low-frequency weight w_{low}	1.0			
	Initial high-frequency weight $w_{\mathrm{high}}^{(0)}$	0.25			
	Final high-frequency weight $w_{\mathrm{high}}^{(1)}$	1.0			
	Spectrum-profile ϵ	10^{-8}			
	Score-space spectrum weighting	Enabled			
Model merging	Full-objective checkpoint	$\lambda_{\mathrm{KL}}\mathcal{L}_{\mathrm{KL}} + \lambda_{\mathrm{GAN}}^{\mathrm{Student}}\mathcal{L}_{\mathrm{GAN}}^{\mathrm{Student}} + \lambda_{\mathrm{GAN}}^{\mathrm{VFM}}\mathcal{L}_{\mathrm{GAN}}^{\mathrm{VFM}} + \lambda_{\mathrm{spec}}\mathcal{L}_{\mathrm{spec}} + \lambda_{\mathrm{HPS}}\mathcal{L}_{\mathrm{HPS}}$			
	Preference checkpoint	$\lambda_{\mathrm{KL}}\mathcal{L}_{\mathrm{KL}} + \lambda_{\mathrm{HPS}}\mathcal{L}_{\mathrm{HPS}}$			
	Merge coefficient α	0.5			
	Final checkpoint	$\theta_{\mathrm{final}} = (1-\alpha)\theta_{\mathrm{full}} + \alpha\theta_{\mathrm{pref}}$			
Inference	Sampling NFEs	1			
	Output space	RGB pixel space			
	VAE decoding	No			
	Throughput protocol	512×512, fp32, 1×H100, 100 samples			
	Throughput	24.42 img/s	24.42 img/s	22.35 img/s	22.35 img/s

Prompt-Only Dataset We combine text prompts from JourneyDB and LAION to construct our prompt-only dataset.

GAN-Supervising Dataset We construct a mixed dataset of real and synthetic images for adversarial supervision. The real images are sourced directly from the LAION dataset. The synthetic images are generated using the teacher model, conditioned on text prompts from JourneyDB.

G Implementation Details

We provide the complete set of training configurations and hyperparameters for our one-step pixel distillation in G.

H Training Algorithm

The complete end-to-end training procedure for our enhanced heterogeneous one-step pixel distillation is detailed in Algorithm 1.

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I Limitations

Our work focuses on one-step generation at 512×512 resolution, with two pixel students and two latent teachers. While this covers both pixel-to-pixel and latent-to-pixel distillation, it does not fully characterize heterogeneous distillation across all architectures, resolutions, or domains. Scaling to higher-resolution generation and stronger future pixel backbones is an important next step.

Although our model is VAE-free at inference, training still uses the teacher VAE encoder to project RGB outputs into the latent teacher space. Our pixel adversarial and spectral objectives reduce this bottleneck, but cannot completely remove biases or missing information introduced by the teacher latent representation.

Our framework also improves inference speed with a more complex training pipeline. The full objective combines latent score distillation, pixel adversarial learning, VFM adversarial supervision, spectral curriculum losses, and preference-calibrated merging. This cost is acceptable for distilling reusable one-step models, but is heavier than naive DMD2.

Finally, the preference-calibrated checkpoint depends on learned reward models such as HPSv2. Although the bounded reward and merging strategy reduce severe reward hacking, reward models remain imperfect proxies for human judgment and can still favor oversharpened or overly saturated images. Like other text-to-image systems, our model may also inherit risks such as biased generation, harmful content synthesis, and misuse for deceptive media.

J More Qualitative Results

We provide additional qualitative results from our one-step pixel model in figs. 7 to 10.

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Figure 7: Qualitative results of our enhanced heterogeneous distillation from FLUX.2-Klein-4B to PixelGen.

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Algorithm 1 Training Algorithm for Enhanced Heterogeneous One-Step Pixel Distillation

Require: pretrained latent teacher T_ϕ; frozen VAE encoder $\mathcal{E}$ and decoder $\mathcal{D}$; real image-text dataset $\mathcal{D}_{\text{real}}$; prompt dataset $\mathcal{Y}$; one-step pixel generator G_θ; fake score estimator T_ψ; student-based pixel discriminator D_{student}; decoupled VFM discriminator D_{VFM}; number of frequency bands K; curriculum length N; spectral weights $w_{\text{low}}, w_{\text{high}}^{(0)}, w_{\text{high}}^{(1)}$; loss weights $\lambda_{\text{KL}}, \lambda_{\text{GAN}}^{\text{Student}}, \lambda_{\text{GAN}}^{\text{VFM}}, \lambda_{\text{spec}}$; preference parameters $\lambda_{\text{HPS}}, \tau_{\text{HPS}}, \alpha$

Ensure: one-step pixel generator $G_{\theta_{\text{final}}}$

Initialize G_θ from a pretrained multi-step pixel diffusion model
Initialize $T_\psi \leftarrow T_\phi$ ▷ fake score estimator
Initialize D_{student} from the pixel diffusion backbone with trainable heads
Initialize frozen VFM backbones $\phi_{\text{SigLIP}}, \phi_{\text{DINO}}$ and trainable heads $g_{\text{sem}}, g_{\text{str}}, g_{\text{fuse}}$ for D_{VFM}
for iteration $\tau = 1$ to max_iters **do**
  Sample $(x_{\text{ref}}, y) \sim \mathcal{D}_{\text{real}}$ and $\xi \sim \mathcal{N}(0, I)$
  $x_0 \leftarrow G_\theta(\xi, y)$ ▷ one-step RGB generation
  $z_0 \leftarrow \mathcal{E}(x_0)$ ▷ project RGB output to teacher latent space
  Sample timestep t and noise $\epsilon \sim \mathcal{N}(0, I)$
  $z_t \leftarrow \text{Forward}(z_0, t, \epsilon)$ ▷ Forward formula in Eq. 1 or Eq. 3
  ▷ **Spectral curriculum weights**
  $\rho(\tau) \leftarrow \min(\tau/N, 1)$
  $h(\tau) \leftarrow w_{\text{high}}^{(0)} + \rho(\tau)\big(w_{\text{high}}^{(1)} - w_{\text{high}}^{(0)}\big)$
  Compute band weights $\{\omega_k(\tau)\}_{k=1}^K$ ▷ Eq. (15)
  ▷ **Spectrum-weighted score gradient**
  $s_{\text{real}} \leftarrow T_\phi(z_t, t, y), \quad s_{\text{fake}} \leftarrow T_\psi(z_t, t, y)$
  $s_{\text{real}}^{\text{adj}} \leftarrow W(s_{\text{real}}), \quad s_{\text{fake}}^{\text{adj}} \leftarrow W(s_{\text{fake}})$
  Compute $\mathcal{L}_{\text{KL}}^{\text{spec}}$ using $s_{\text{real}}^{\text{adj}} - s_{\text{fake}}^{\text{adj}}$ ▷ Eq. 19
  ▷ **Spectrum-profile loss**
  $\hat{z}_{0,\phi} \leftarrow \text{CleanEstimate}(T_\phi, z_t, t, y)$ ▷ Clean prediction formula in Eq. 3
  $x_{\text{teacher}} \leftarrow \mathcal{D}(\hat{z}_{0,\phi})$
  Compute $\mathcal{L}_{\text{spec}}(x_0, x_{\text{teacher}})$ with $\{\omega_k(\tau)\}_{k=1}^K$ ▷ Eq. 17
  ▷ **Student-based pixel generator update**
  Construct noisy RGB images $x_{0,t}$ and $x_{\text{ref},t}$ using the forward process ▷ Forward formula in Eq. 1 or Eq. 3
  $\mathcal{L}_{\text{GAN},G}^{\text{Student}} \leftarrow f(-D_{\text{student}}(x_{0,t}, t, y))$
  $\mathcal{L}_{\text{GAN},G}^{\text{VFM}} \leftarrow f(-D_{\text{VFM}}(x_0, y))$
  ▷ **Full-Objective for Generator**
  $\mathcal{L}_G \leftarrow \lambda_{\text{KL}}\mathcal{L}_{\text{KL}}^{\text{spec}} + \lambda_{\text{GAN}}^{\text{Student}}\mathcal{L}_{\text{GAN},G}^{\text{Student}} + \lambda_{\text{GAN}}^{\text{VFM}}\mathcal{L}_{\text{GAN},G}^{\text{VFM}} + \lambda_{\text{spec}}\mathcal{L}_{\text{spec}}$
  $G_\theta \leftarrow \text{update}(G_\theta, \mathcal{L}_G)$
  ▷ **Fake score estimator update**
  Sample timestep t' and noise $\epsilon' \sim \mathcal{N}(0, I)$
  $z_t' \leftarrow \text{Forward}(\text{stopgrad}(z_0), t', \epsilon')$ ▷ Forward formula in Eq. 1 or Eq. 3
  Compute $\mathcal{L}_{\text{fake}}$ for T_ψ ▷ Eq. 2 or Eq. 4
  $T_\psi \leftarrow \text{update}(T_\psi, \mathcal{L}_{\text{fake}})$
  ▷ **Student-based pixel discriminator update**
  $\mathcal{L}_{\text{GAN},D}^{\text{Student}} \leftarrow f(D_{\text{student}}(x_{\text{ref},t}, t, y)) + f(-D_{\text{student}}(\text{stopgrad}(x_{0,t}), t, y))$ ▷ Eq. 8
  $D_{\text{student}} \leftarrow \text{update}(D_{\text{student}}, \mathcal{L}_{\text{GAN},D}^{\text{Student}})$
  ▷ **Decoupled VFM discriminator update**
  Compute SigLIP and DINOv2 logits for x_{ref} and x_{x0} ▷ Eq. 9 to Eq. 10
  Compute $D_{\text{VFM}}(x, y) \leftarrow g_{\text{fuse}}([\ell_{\text{SigLIP}}(x, y); \ell_{\text{DINO}}(x)])$
  Compute $\mathcal{L}_{\text{GAN},D}^{\text{VFM}}$ on (x_{ref}, y) and $\text{stopgrad}(x_0)$ using Eq. 12
  Update only $g_{\text{sem}}, g_{\text{str}}, g_{\text{fuse}}$
end for
$\theta_{\text{full}} \leftarrow \theta$ ▷ full-objective checkpoint
▷ **Preference-calibrated checkpoint and merging**
Train θ_{pref} using $\mathcal{L}_{\text{KL}} + \lambda_{\text{HPS}}\mathcal{L}_{\text{HPS}}$, where $\mathcal{L}_{\text{HPS}} = \max(0, \tau_{\text{HPS}} - R_{\text{HPS}}(x_0, y))$
$\theta_{\text{final}} \leftarrow (1 - \alpha)\theta_{\text{full}} + \alpha\theta_{\text{pref}}$
return $G_{\theta_{\text{final}}}$

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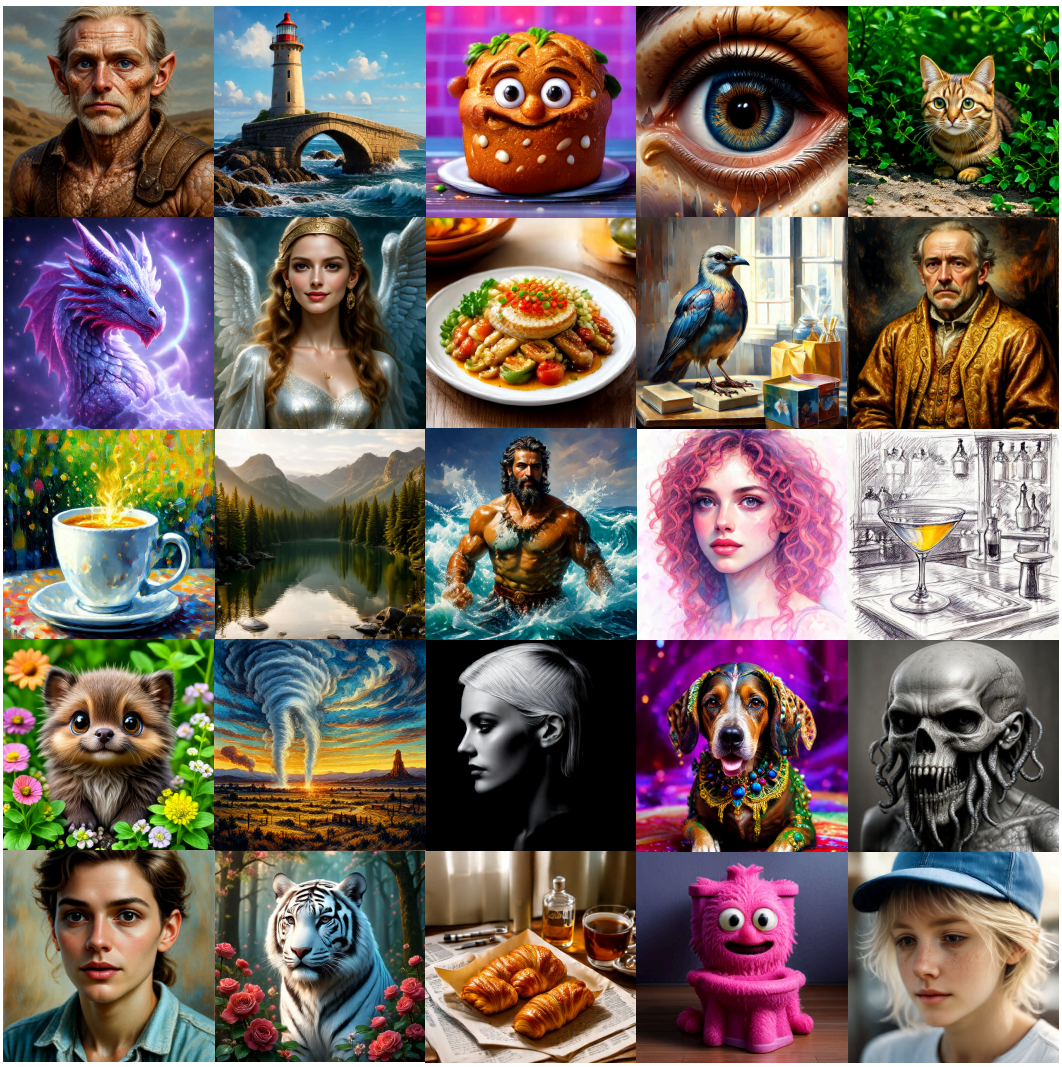

Figure 8: Qualitative results of our enhanced heterogeneous distillation from SD 3.5 Medium to PixelGen.

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Figure 9: Qualitative results of our enhanced heterogeneous distillation from FLUX.2-Klein-4B to PixNerd.

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Figure 10: Qualitative results of our enhanced heterogeneous distillation from SD 3.5 Medium to PixNerd.

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   Justification: The abstract and introduction state the main contribution as heterogeneous score distillation from latent teachers to one-step VAE-free pixel students. The claims are supported by the quantitative comparisons, ablations, qualitative results, and inference-efficiency analysis.

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